

TDT4175 - INFORMATION SYSTEMS

Enhancing NTNU's Teaching Processes through Generative AI: An Analysis and Implementation Proposal

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Table of Contents

List of Figures	ii
1 Executive Summary	1
2 Background	2
2.1 Information Efficiency & Accessibility	2
2.2 Online Tools for Education	4
2.3 Disadvantages	5
3 Teaching at NTNU: the AS-IS situation	6
3.1 Assignment Creation	7
3.2 Group formation	8
3.3 Grading assignments	9
4 Teaching at NTNU: the TO-BE situation	12
4.1 Assignment Creation	12
4.2 Group Formation	14
4.3 Grading assignments	16
5 From AS-IS to TO-BE — Implementation	18
5.1 Implementation plan	18
5.1.1 Managing change	18
5.1.2 Implementation strategies	19
5.1.3 Choosing a strategy	20

5.2	Risk and management strategies	22
5.2.1	Resistance	22
5.2.2	Security	23
Bibliography		25

List of Figures

1	BPMN: Teaching at NTNU AS-IS	6
2	BPMN: Assignment creation AS-IS	8
3	BPMN: Group Formation AS-IS	9
4	BPMN: Grading assignments AS-IS	11
5	BPMN: Assignment creation TO-BE	14
6	BPMN: Group formation TO-BE	15
7	BPMN: Grading assignments TO-BE	17

1 Executive Summary

The purpose of this report is to examine the educational system and its processes, at Norwegian School of Science and Technology (NTNU). The focus of the report is on how the use of Information and Communication Technologies (ICT) can further develop and innovate the teaching process. The report starts by describing the evolution of ICT, and highlighting its current and future advancements.

Bearing these ICT advancements in mind, the report moves on to describe the current AS-IS situation of the teaching process at NTNU (See figure 1). The overall teaching structure, as well as different sub-processes, are described using the Business Process Model and Notation (BPMN) model. There are models for both the AS-IS situation and the TO-BE situation, which incorporates the group's proposed improvements. The BPMN describes the current teaching process at NTNU, first outlining the different actors, including the lecturer, teaching assistant (TA) and the various administrative parts of NTNU.

Furthermore, this report will focus on the sub-processes regarding assignment creation, group formation and assignment grading. Considering these sub-processes, we will examine how Artificial Intelligence (AI) can be used to possibly improve these systems. The BPMN models will be used to learn where AI potentially may fit into all these processes, having the end-goal of reducing the workload of our core actors.

The report concludes by outlining a structured implementation plan for transitioning from the AS-IS to the suggested TO-BE situation. Describing potential strategies for managing this organizational change, discusses associated risks, particularly regarding AI in education, and emphasizes the crucial role of NTNU's administrative and communications support in ensuring successful implementation.

2 Background

The educational system has in the past few years changed dramatically, both in the way the lecturer teach, how students learn and the tools that are accessible for educational purposes. Information and communication technologies (ICT), is a huge part of the drastic change in education. The use of ICT can lead to higher order of thinking skills, increase creativity, and the options that the student have to learn and express their understanding [4]. However, the introduction of these ICT tools must be carefully considered, discussing how they should be implemented and how they will affect the education.

2.1 Information Efficiency & Accessibility

One of the most significant impacts of ICT on education has been the transformation of communication between schools and students. In the past, information was often only available through oral lectures or directly from professors, making it difficult for students who were unable to attend classes. Schools today provide digital tools for communication, through for example E-mail and Blackboard¹. This allows students and professors to exchange information from different locations and at different times. These changes to the academical structure gives the students more flexibility and control over their education.

The overall structure of the education has seen some drastic changes, with its transition from chalkboards to electronic devices. This has expanded learning possibilities for both students and lecturers. Even notebooks has been replaced by devices such as iPad's or digital notebooks, like Remarkable, which offer an opportunity for handling class notes in a more organized manner.

This use of electronic devices also add a high potential for improving the student education by providing more creativity and visual thinking, which has been shown to improve student learning outcomes. Visual thinking has been supported by researches as a method that helps with processing the course material [6, 9]. With

¹An educational platform for information exchange.

the use of electronic devices, student can add images, graphs and clip-outs from the lecturer notes, to add a deeper level of understanding and making the notes more useful and easy to remember.

For lecturers, the use of these devices also bring multiple benefits. Using digital devices when holding lectures may help with the students' attention, and making the lectures more engaging. Another benefit is the option to share the digital lecture notes, allowing students to come back to the material when needed, facilitating for students to work at their own pace.

Furthermore, digital tools allow for the automation of time-consuming and repetitive tasks. This, in turn, may further increase the efficiency of the lecturing process and the educational system. For example, the introduction of automated grading and exam administration has seen a rise in the past few years. This may be further enhanced either by designing tests that a computer can automatically grade, or by using other tools.

The change in the educational system has also improved the educational accessibility. The use of online webinars, or free courses created by universities such as Massachusetts Institute of Technology (MIT), have globalized educations. Students can now attend their education remotely, or learn more about a subject with the use of internet.

Many of these new systems became very important during the Covid pandemic, as this period led to testing and adapting of many new online tools and systems. Schools changed to online lectures using platforms such as Microsoft Teams or Zoom to continue educating student, even when physically lectures were restricted. While the development of new educational methods was important, other aspects of the school system such as tests and exams also required adaptation.

Online tests allowed students to complete assessments from home. To ensure academic integrity and prevent cheating, new systems were developed or updated to accomplish this task. One example is Safe Exam Browser, which restricts access to external websites during an exam. In some cases, traditional written exams were replaced by oral exams over different platforms, offering a more secure and interact-

ive way to evaluate students remotely. These innovations changed how education is conducted in the present, and show a possible future in the educational system.

2.2 Online Tools for Education

Educational tools have always played a crucial role in learning. Traditionally, physical instruments like drawing compasses and protractors helped students visualize and understand different subjects. In the modern day, these tools are still used, but new opportunities have emerged with the rise of the internet. Online tools such as GeoGebra, Kahoot and Mentimeter have been more and more introduced in today's education. These tools enhance the student's understanding of the subject through hands on visualizations, but also introduce the new concepts, such as gamification.

Gamification help students with more than learning the material, it encourages student interaction, collaborations and communication. Students can work together to solve tasks, creating experiences that are just as important as learning the material itself [1].

As mentioned, the use of online tools provides an unrivaled flexibility in accessing learning material. Platforms such as YouTube is often used for entertainment, but contains an assortment of videos for learning. *3 Blue 1 Brown* is one of these channels, which provides visual learning of mathematics. *Khan academy* is another example, which provides free coding tutorials online. This flexibility provided by such tools allow the student to learn at their own pace by pausing, skipping or replaying videos, further introducing educational freedom.

Furthermore, digital tools open for real-time collaboration over the web. For example, Google Drive, Jira or even CodeWithMe makes it easy to share ideas, and to co-create projects without the need to be in the same room. This doesn't only give the students a new way of solving different tasks, but also introduce the students in nature of regular work environments.

2.3 Disadvantages

While the evolution of ICT's offers many improvements to the existing educational system, it also brings several disadvantages. One important issue is the introduction of Artificial Intelligence (AI) technologies. Even though AI can be used to improve system efficiency and organizational structure, it can lead to misuse.

AI is often used to lighten student workload by AI-generated articles or coursework. Due to the exponential improvement in AI technology, this misuse can create work indistinguishable from human-written work. Tools such as ChatGPT can therefore be used unethically, for example with the intention of plagiarism. The use of such tools can also lead to misinformation, where AI creates wrong facts and fake articles. Additionally, the use of AI creates a concern about academic integrity. Even though plagiarism checks and anti-cheat tools are improving, they can also lead to false accusations where actual human-written text is incorrectly flagged as AI-generated.

Another disadvantage with the evolution of ICT's is the rapid shift to online education. Since school isn't only about education, the shift towards remote education can lead to students under-developing important soft skills that are often learned in face-to-face education and interaction. Which is even more true for younger students. These are skills such as communication and peer interaction. A scientific article about Covid and school as a socialization place noted that over half of the participants expressed anticipation to return to school, mainly for the social interactions that in-person schooling provides [5, p. 5-8]. Without the physical school environment, students may only have virtual interactions, leading to isolation [7, p. 259].

3 Teaching at NTNU: the AS-IS situation

This section will discuss the current teaching methods at NTNU and provide an overview using BPMN modeling. Three important sub-processes within the teaching methods will be further explained and evaluated. These are assignment creation, group formation, and grading of assignments.

The AS-IS teaching process at NTNU varies significantly between courses. Some are conducted entirely digitally or physically, but today, most adopt a hybrid approach. From a lecturer's perspective, the general structure involves planning the course, holding lectures, and planning the exam/deliverable. Most courses also have assignments throughout the semester that must be passed to qualify for the exam or to submit a final deliverable. Additionally, most courses have teaching assistants (TAs) to assist the lecturer in several areas, such as helping students and grading assignments.

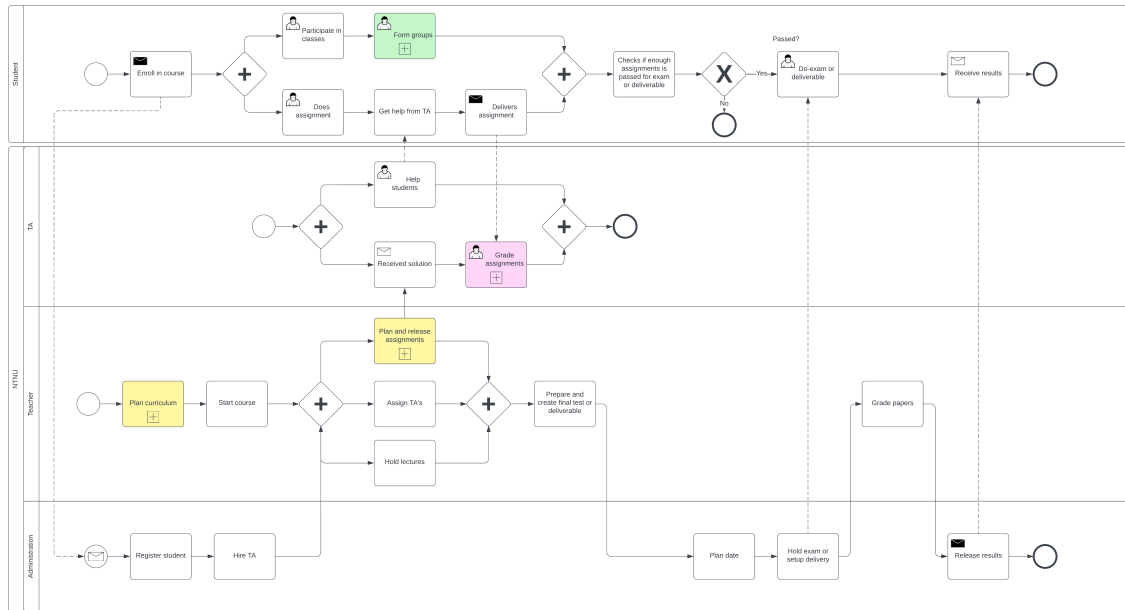


Figure 1: BPMN: Teaching at NTNU AS-IS

The overview model, illustrated in Figure 1, presents a simplified AS-IS teaching process at NTNU. Key roles include lecturers, TAs, students, and administration. The model shows the flow from enrolling in a course to the examination, while emphasizing the grading of assignments, group formation, and planning of courses.

The model is created based on a student's perspective.

3.1 Assignment Creation

At NTNU, lecturers are responsible for creating and managing course exercises, and this process remains largely manual. The workflow begins with the lecturer deciding whether to create new assignments or reuse assignments from previous years. If reusing, the lecturer retrieves past assignments from the assignment platform and adapts them as needed.

When creating a new assignment, the lecturer reviews the course syllabus to ensure alignment with the course's learning objectives. The assignment is then manually written, including any instructions for students to complete. Additionally, the lecturer creates a solving proposal, which outlines the expected solution and serves as a guide during the grading process. This solving proposal is essential for ensuring grading consistency, particularly when teaching assistants (TAs) are involved.

After completing the assignment and solving proposal, the lecturer sets a submission deadline. This deadline is manually entered into platforms like Blackboard, Eduflow or similar, ensuring students are aware of when the assignment is due. The lecturer also sets the publishing date, controlling when the assignment becomes visible to students and TA's.

Although functional, this entire process is mostly manual, which is time-consuming and requires significant effort from the lecturer to creating new assignments. Creating assignments that align closely with the syllabus and developing solving proposals manually can be a challenging, especially when aiming for consistency and quality across courses.

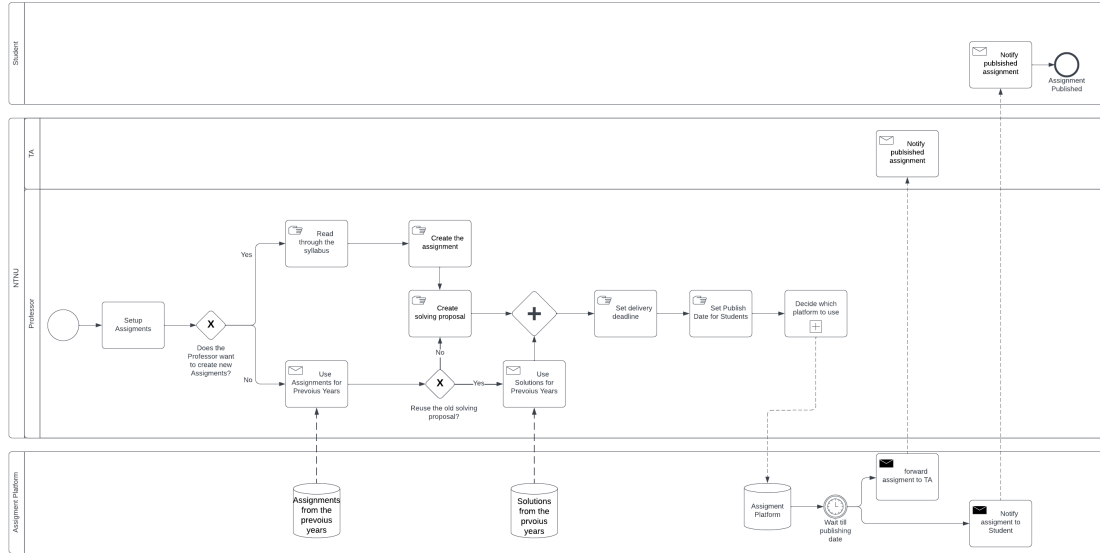


Figure 2: BPMN: Assignment creation AS-IS

3.2 Group formation

Many courses at NTNU include some type of group-exercise to be done during the semester. This could include mandatory assignments that must be completed to qualify for the exam, or a final group project as the course's main deliverable.

The lecturer is responsible for setting the structure for group work, determining details such as the size of each group and the total number of groups required based on the number of students enrolled. This planning is essential, as group assignments often form a crucial part of the coursework, and ensuring an even distribution of students across groups is necessary for balanced collaboration.

Once the framework for group formation is established, the lecturer creates the group slots within the Blackboard system. This allows students to sign up for groups, either with peers they have already connected with or individually, if they have not yet found a complete group. While some students may organize themselves in advance and complete the process quickly, others rely on the lecturer to finalize their group, particularly if they are new to the course or haven't formed connections with enough classmates.

Once students have signed up, the lecturer reviews the group compositions, identi-

fying any groups that are missing members. At this stage, the lecturer allocates the remaining students to ensure that every group has the appropriate number of members, balancing the distribution of students. This part of the process requires the lecturer to consider factors such as the total number of students, and also ensure that no group is too small, providing a more balanced learning experience for all.

Over the course of the semester, there may be instances where groups lose members due to students dropping the course or other unforeseen circumstances. When this happens, the lecturer must reassess the group compositions and reassign students if necessary, ensuring that the remaining students are not left at a disadvantage by losing a member. This ongoing oversight by the lecturer is crucial for maintaining group stability and helping students succeed in their collaborative efforts, regardless of any challenges that arise during the course.

This process requires a lot of manual work and oversight from the lecturer of the course, both in the beginning of the course and throughout the semester.

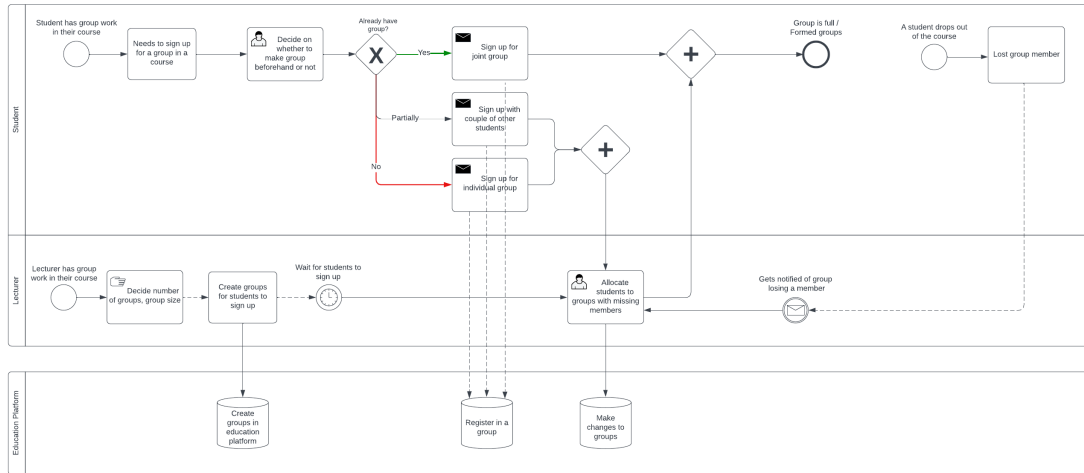


Figure 3: BPMN: Group Formation AS-IS

3.3 Grading assignments

The grading of assignments is considered a crucial aspect of the teaching process at NTNU. Mandatory assignments have become an essential component in most courses, often serving as milestones for both students and lecturers . Consequently,

providing students with feedback is vital to help them stay on track. Moreover, students must pass a set number of assignments to be eligible for the exam or to submit a final deliverable.

While grading practices vary significantly between courses, the group’s experience indicates that Teaching Assistants (TAs) often handle grading and feedback. If a student fails an assignment, they are typically given the opportunity to redo it, usually with the lecturer’s involvement. This process was deemed important to model due to its time-consuming nature, especially considering that some courses have several hundred students distributed on very few TA’s. To identify potential areas for improvement, a BPMN model of this process was created.

The sub-process *Grading assignments* is illustrated in Figure 4. As mentioned, TAs are expected to handle the grading of assignments. In real-world scenarios, several platforms are used for grading: the *Information systems* course uses Eduflow, mathematics often uses Capquiz, while others use Blackboard. This model is simplified to consider only Blackboard, disregarding other grading platforms. Ideally, all grading would occur on a single platform; however, this is not feasible due to the wide variety of potential assignments.

The modeled process begins with the release of an assignment and repeats each time a new assignment is released. When a student submits the assignment on time, the TA starts reviewing and grading. The lecturer is expected to provide both a rubric and a solution set, which the TA uses to assign a score and provide feedback. After a score is set, it is compared to the required score, resulting in either a pass (forwarded to the student) or a fail (forwarded to the lecturer). The lecturer then decides whether to allow the failed student another attempt.

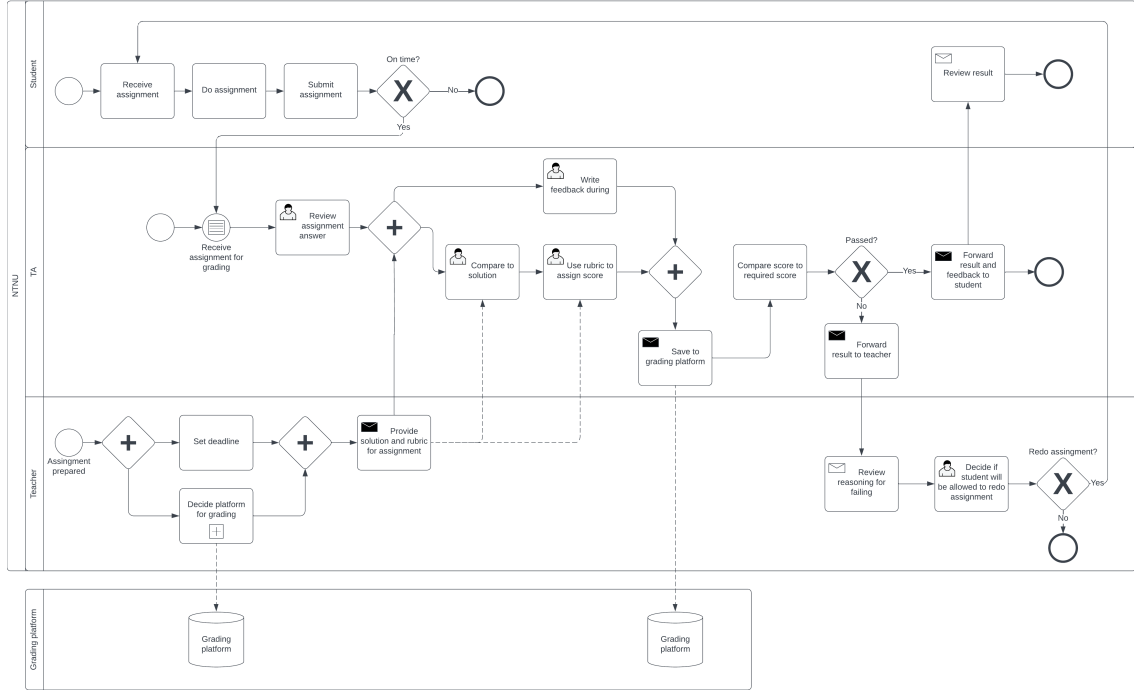


Figure 4: BPMN: Grading assignments AS-IS

This section has covered the AS-IS situation at NTNU, focusing on assignment creation, group formations and assignment grading. All models for the AS-IS is based on student's perspectives, and is biased and also speculative. Using the same definition of all courses is not natural, but does provide a generalized view of courses taught at NTNU. Today, assignments are created by lecturers, with options to reuse previous assignments if they exist. This process is largely dependent on the lecturer's preference. Group formation is mostly a student task, while involving the lecturer in the event of missing group members or students not finding a group. This is highly dependent on the course as well. Grading of assignments we concluded were the task of a TA, while this has been true on many of the group's courses, it may differ from courses. It is still dependent on the lecturer's influence, with TAs requiring rubric and solution for grading. Improvements to these using generative AI will be discussed in Section 4.

4 Teaching at NTNU: the TO-BE situation

This section will discuss potential improvements to the sub-processes described in the previous chapter, specifically focusing on how artificial intelligence can be implemented to increase efficiency and reduce the workload of some of the staff at NTNU.

4.1 Assignment Creation

A challenge in the current assignment creation process at NTNU is the high degree of manual labor required for lecturers to design, create, and manage assignments. The process relies heavily on the involvement of the lecturer at every stage. The proposed TO-BE situation introduces Generative AI (GenAI) to automate key parts of this process, ensuring a more efficient workflow while maintaining the lecturer's authority over final content decisions.

In the TO-BE model, the BPMN diagram 5 illustrates how this integration of GenAI into assignment creation streamlines the workflow. The system automates the generation of assignments and solving proposals, freeing up time for lecturers to let them focus on higher-value tasks like content refinement and student interaction.

As in the AS-IS workflow, the process begins when the lecturer decides whether to create new assignments or reuse existing ones. If the latter is chosen, the system retrieves past assignments and corresponding solving proposals from the *grading platform*. This ensures that previously successful assignments can be reused with minimal effort.

When the lecturer chooses to create a new assignment, the system requires input in the form of assignment criteria. This is a manual task where the lecturer provides guidance for what kind of assignment is expected. Once these criteria are entered, the GenAI system takes over and generates both the assignment suggestions and the corresponding solving proposals.

The assignment suggestions and solving proposals are then reviewed by the lecturer. At this stage, decision points, labeled "Satisfied with the Assignment Suggestion?" and "Satisfied with the Solving Proposal?", are used to determine whether the content meets the lecturer's expectations. If the suggestions do not align with the objectives, feedback may be provided for improvements, and GenAI will regenerate the suggestions based on this input. This creates an iterative feedback loop, allowing lecturers to refine the assignment content while still benefiting from the efficiency of AI automation.

Once the lecturer is satisfied with both the assignment and solving proposal. The process shifts to finalizing practical details, such as setting the assignment deadlines, publish dates and platform decision, mirroring the «AS-IS». These tasks remain manual, as they require the lecturer's judgment based on course timing and other factors.

By introducing this TO-BE process, NTNU stands to gain good improvements in the efficiency of assignment creation. Lecturers will experience a reduction in manual tasks, especially during the content generation phase, while still maintaining the ability to control and fine-tune the assignments before they are released to students. Additionally, by automating assignment generation, lecturers will be able to introduce greater variety in assignments year-to-year, ensuring that course content is renewed more often and students are exposed to fresh challenges each term.

To further improve the assignment creation experience, the use of GenAI in this process could incorporate additional features such as adaptive learning mechanisms or personalized assignments based on individual student performance data. This would ensure that the assignments are not only relevant to the course but also tailored to individual learning needs, which could be a powerful step forward in enhancing NTNU's educational offerings.

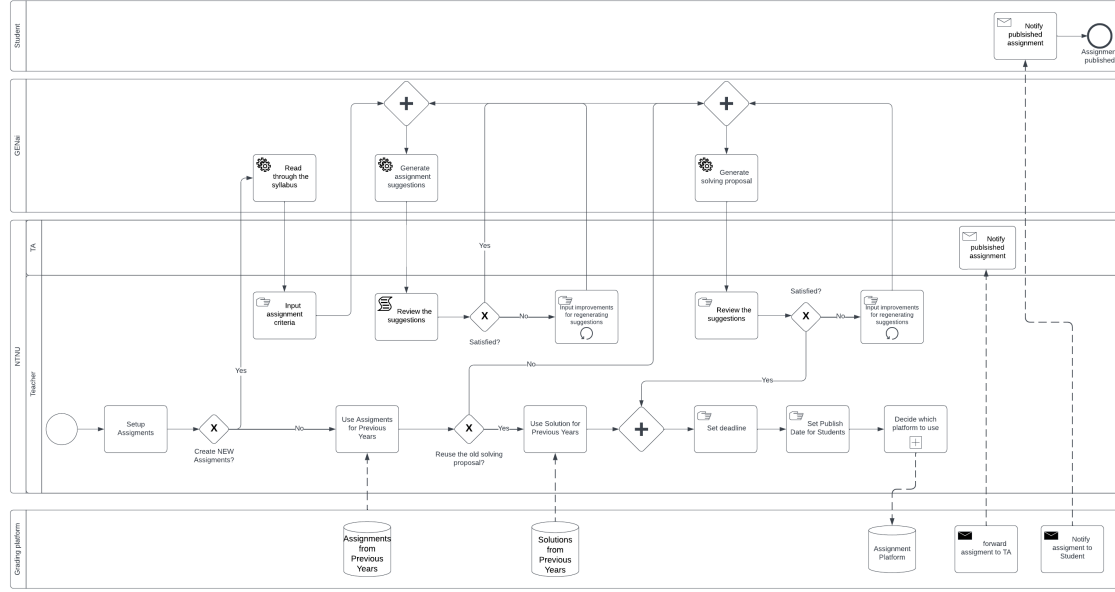


Figure 5: BPMN: Assignment creation TO-BE

4.2 Group Formation

The current group formation heavily involves the lecturer having to delegate students into groups in the cases where groups are not full. It is specifically this process that the group wanted to highlight as the main problem in the AS-IS model. If there are many students in the course, the delegation process can be very time-consuming for the lecturer. This time could be more effectively used for other course-related work. In addition, especially in college courses, the lecturer does not know the students and their work habits or ambitions for the course, which in turn makes it even more challenging to create fair and equal groups.

The new TO-BE model, as seen in Figure 6, focuses on easing the workload for the lecturer, by having generative AI create the groups in the cases where the students either does not have a group, or the group is not filled. The new model begins with the lecturer publishing a questionnaire that contains questions about workload and expectations about the course work that each student has to fill out. If a student already has a full group, they can sign up for the joint group in Blackboard as normal. However, if a student does not have a full group, then the generative AI receives the response from the questionnaire, as well as previous grades that the

student has achieved, from StudentWeb. Hence, the generative AI creates groups based on the received data. After the groups are formed, the lecturer has the chance to review the formed groups and can either approve or reject the group formation. If, for some reason, the lecturer chooses to reject the group formation, the generative AI regenerates the groups again until the groups get approved. After the formed groups are approved, the lecturer registers the groups to Blackboard, as well as notifies the students of their new group members, e.g. through an announcement in Blackboard.

In comparison to the AS-IS model, the TO-BE model reduces the workload for the lecturer by not having to manually create groups. The only additional tasks that the lecturer has to prepare compared to the AS-IS model, is creating a standardized questionnaire, as well as reviewing the groups formed by the generative AI. This approach also benefits the students because it connects students with equal expectations and ambitions for the course, as well as provides an opportunity to create balanced groups in terms of fair workload distribution among the group members.

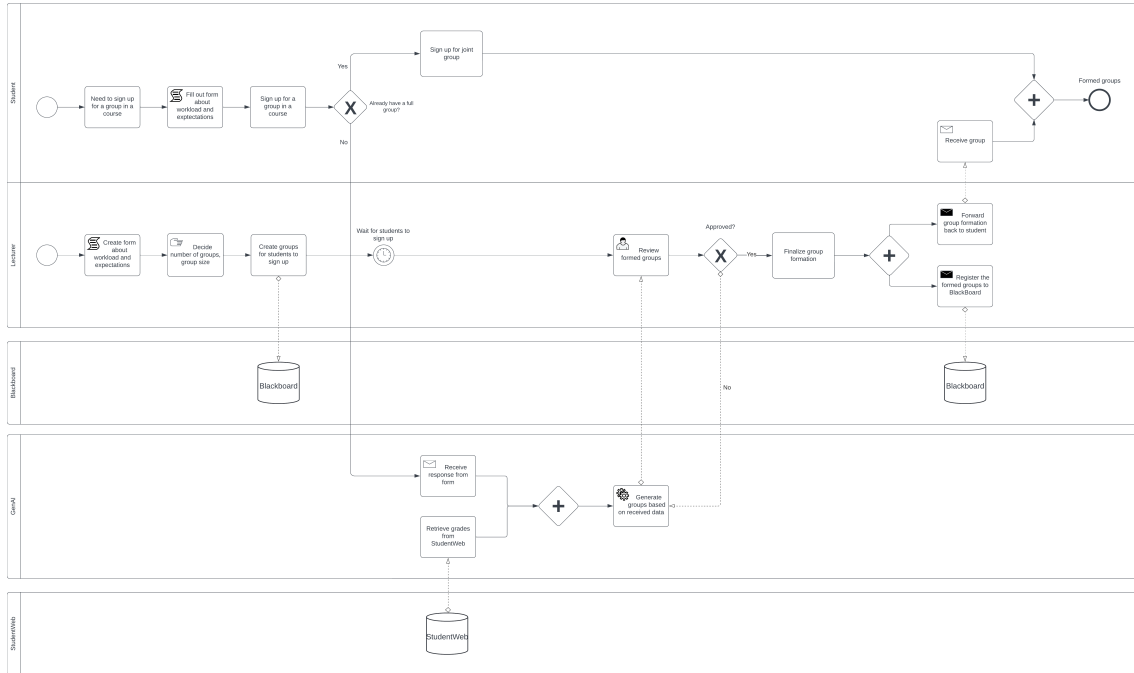


Figure 6: BPMN: Group formation TO-BE

4.3 Grading assignments

The grading process at NTNU currently involves multiple actors, platforms, and inconsistent approaches to feedback. For instance, lecturers often rely on various platforms, which may lead to inefficiencies. Often the primary reason of using some platforms is the familiarity with a specific platform. The proposed TO-BE scenario aims to streamline this by centralizing grading within a unified platform that integrates both automated and manual grading procedures. As noted by Stair et al. (2021), it is essential that valuable information be readily accessible to enhance overall efficiency [8, p. 8].

In the envisioned TO-BE model, students submit their assignments to a single grading platform integrated with GenAI that performs automated checks. These checks include plagiarism detection, basic compliance with submission rules, and in some cases, basic content analysis using natural language processing (NLP). For example, AI algorithms can scan for missing citations, incorrect referencing, or structural inconsistencies. Automation at this stage serves to reduce the workload on human graders by flagging potential issues early in the process. If an assignment passes the automated checks, it is then forwarded to TAs for manual review. Stair et al. (2021) emphasizes that implementing automated systems for initial review tasks increases both efficiency and accuracy in workflows, which may enable graders to focus on content quality and critical analysis [8, p. 8].

The streamlined feedback loop is maintained. After manual grading, students receive detailed feedback outlining areas of strength and improvement. If the course policy allows, students may resubmit their work after addressing the feedback². The feedback loop is designed to foster a learning environment where students can improve based on constructive criticism. According to Dewey (1938), such iterative feedback systems help students engage with content more deeply, encouraging critical thinking and problem-solving [2, p. 5-8].

By employing a centralized grading platform, it replaces the need for multiple systems for submission and grading. As highlighted by Stair et al. (2021), centralized

²It is assumed that students may not resubmit assignments if the deliverable is plagiarized.

systems improve information accessibility, making it easier for students to track their submissions and grades, and for educators to manage their workloads [8, p. 8]. This platform could integrate with existing university systems (e.g., Blackboard, Edulflow).

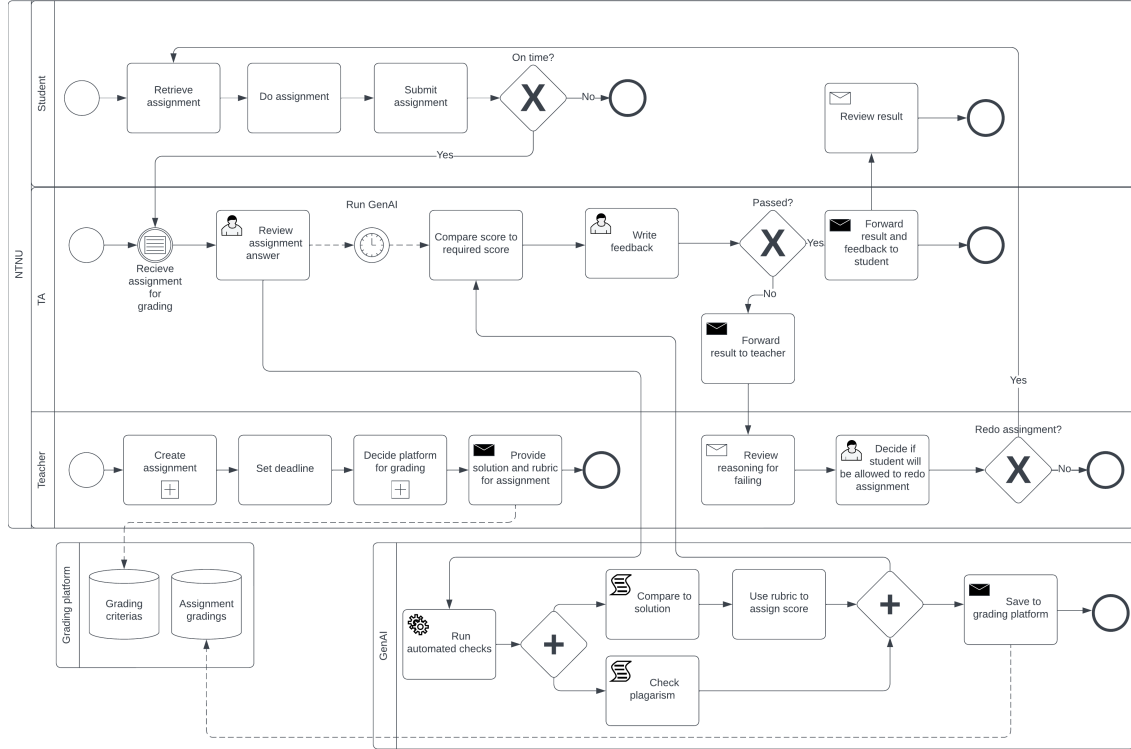


Figure 7: BPMN: Grading assignments TO-BE

5 From AS-IS to TO-BE — Implementation

Shifting from the current AS-IS situation to the suggested TO-BE situation requires a structured and well-thought-out implementation plan. This section will describe this plan, as well as discuss the main risks that could interfere with a successful use of this proposed solution. These are risks primarily regarding resistance and security, and we will discuss both their cause and how to deal with them.

5.1 Implementation plan

As we can see from the described AS-IS processes, the teachers have to do a lot of manual work regarding assignment creation, group formation and grading assignments. The suggested TO-BE situation will help reduce the workload by using AI to improve these processes, making them more efficient and optimizing the outcome. Given the implementation plan, it is essential to consider how to effectively manage the upcoming changes in the system. Equally important is evaluating various implementation strategies, discussing their pros and cons, and determining the most suitable approach for the implementation in each sub-process.

5.1.1 Managing change

In the purposed curriculum for this course, it is stated that "The ability to manage change is critical to the success of systems development" [8, p. 383]. New systems inevitably cause change, regardless of the magnitude. An important part of managing change is therefore to recognize important aspects of the new system, and especially the potential problems that it can bring forth.

After the recognition of the problems, it is important to deal with them before they escalate. This is particularly important for problems related to users, in this case the students, teachers and teaching assistants [3] [8, p. 383]. Following the most common problems mentioned by Stair et al. (2021), it is essential to consider the issues in the context of the purposed changes of this report.

One potential concern is the *fear of losing job security and influence*. The likelihood of this occurring is relatively low, as Generative AI will not replace but rather simplify manual work. However, as the fear of job loss related to AI grows, it can be a concern for some of the users, especially the teaching assistants and their work related to grading of assignments. Communication is therefore an important factor in reducing this problem, along with providing information, and advocating for the change. Effective leadership should clarify the changes, their factors, the timeline for the implementation, and how they will impact the current system. Additionally, gathering valuable input from the users will ensure a high standard of the transition.

Another problem that could occur is that the change will *negatively affect the structure and create more work than the old system*. As the use of Generative AI in educational systems is rather new, there will inevitably be a learning curve as the users adapt to the new system. During this phase, it is therefore important to still mainly focus on the user and be ready to receive user feedback through NTNU's channels. Additionally, it will be important to ensure reliable IT support so that users can perform their work and solve technical issues that arise. To minimize the problem of system change, communication is still key, as unclear expectations and understanding of the new system will lead to problems. Organizations such as NTNU could benefit by timely planning the implantation of the new system, with the best possibility being during the summer break, as it's the time when users are less reliant on the system.

Unfortunately, not everyone manages change easily, which can in some cases increase their problems, and the complexity of the system. As humans are different, some difficulties and problems are inevitable.

5.1.2 Implementation strategies

There are many ways to implement the information systems and make them fully operational. This process is called start-up and includes various different implementation approaches. These approaches both have pros and cons that must be carefully considered when implementing the said information system.

Direct conversion is an approach where the old system is stopped and thereafter the new system is started on a given date [8, p. 431]. This approach is generally not preferred because of the high risk of complications occurring when switching systems abruptly without optimizing or changing the new system.

A more preferred and safe approach is called the phase-in approach, which involves slowly phasing in components of the new system while slowly phasing out components of the old system [8, p. 431]. This process is repeated until the new system is fully integrated and operational in all applications. The phase-in approach could potentially be quite time-consuming because of the constant and gradual replacement of the components, which is an important factor to keep in mind.

Pilot running is an approach that utilizes direct conversion to a group of users, rather than all the users [8, p. 432]. It involves implementing a pilot to a chosen user group, and if it is successful it can be implemented to another user group. This ensures that the information system is implemented gradually and tested within a part of a bigger environment.

The final approach is called parallel running and is a way to run both the old and new system for a period of time [8, p. 432]. This approach involves running both systems simultaneously, and whenever the users feel comfortable with the new system, the old system is disbanded. This way, all users can adapt to the new system gradually without having to immediately switch to a system that they have no experience with.

5.1.3 Choosing a strategy

When moving from the AS-IS to the TO-BE state, various factors must be considered when selecting an appropriate implementation strategy. Each process — assignment creation, group formation, and assignment grading — requires a distinct approach. Ultimately, the goal is to transition efficiently while minimizing risks and disruptions.

For assignment creation, the Direct Conversion strategy is the most suitable. Currently, teachers rely on various platforms to create and distribute assignments, often

due to their familiarity with these systems. This results in inconsistent practices and inefficiencies, as seen in the AS-IS model, where different systems lead to variations in assignment formats. In the TO-BE scenario, having a unified platform will ensure that all assignments are created, distributed, and tracked consistently across all courses. As Stair et al. (2021) point out, direct conversion can be effective in forcing users to adapt quickly by eliminating the comfort of the old systems [8, p. 431]. Since the assignment creation process does not necessitate keeping the old systems running concurrently, a direct, immediate transition will ensure uniformity without the complications of managing two systems at once.

In contrast, for group formation, a more cautious Pilot Running strategy is preferable. Group formation involves balancing complex factors such as student workloads, performance, and preferences. There is also a significant risk of dissatisfaction if the AI-driven system does not function as expected. This makes it essential to test this system on a smaller scale before full-scale deployment. By piloting the group formation tool in a few selected courses or departments, NTNU can gather feedback and make necessary adjustments to the algorithm. This strategy allows for fine-tuning the system based on initial results and avoids the risk of widespread dissatisfaction. Unlike the direct conversion for assignment creation, group formation is more subjective, and there's a need to ensure that the AI handles the complexities of student needs.

Assignment grading is another process that benefits from the Direct Conversion strategy. Similar to assignment creation, grading currently involves multiple platforms and inconsistent approaches, driven by a lot of the same factors. This fragmentation results in inefficiencies and variability in how grades and feedback are delivered to students. Shifting directly to a single, centralized grading platform forces immediate adoption and ensures standardization. As with assignment creation, maintaining dual systems would only extend the inefficiencies of the AS-IS state, and there is no strong need for a transitional period.

It should be noted that Direct Conversion has its challenges. As Stair et al. (2021) highlight, skipping a transition period can leave users unprepared, potentially causing errors during implementation [8, p. 431]. Despite this, the long-term benefits of

standardization for assignment creation and grading outweigh these risks. Parallel Running, which involves running both old and new systems simultaneously, would be unnecessary for these processes, but could be useful for more content-dependent systems. However, group formation benefits from a Pilot Running strategy, where the system can be tested and refined before full-scale implementation. Thus, for these processes, Direct Conversion is ideal for assignment creation and grading, while Pilot Running is better suited for group formation.

5.2 Risk and management strategies

Introducing new technologies, such as Generative AI into established educational workflows at NTNU, involves various risks. These risks need to be carefully managed to ensure a smooth transition from the current AS-IS processes to the proposed TO-BE scenario. Two of the most significant risks in this context are resistance to change and security concerns, particularly in relation to GDPR and data privacy.

5.2.1 Resistance

Resistance to change is a common issue in any organization facing the implementation of new systems. According to Stair et al., not everyone adapts easily to change, especially when the changes introduce new technology that could alter job roles or workflows [8, p. 383]. In the case of NTNU, introducing Generative AI in processes like assignment creation, grading, and group formation may lead to concerns among teachers and teaching assistants (TAs). These concerns often stem from fears of job loss, the perceived complexity of learning new systems, and the potential disruption of existing workflows.

For example, teachers who have long relied on manual processes for assignment creation may feel uncertain about the accuracy or quality of AI-generated content. Such concerns can slow the adoption of AI systems and limit their effectiveness.

To address these concerns, it is essential to implement effective change management strategies. Communication will be key in this process. By clearly explaining how

AI is intended to support, rather than replace. AI should be framed as a tool that reduces manual workload and allows staff to focus on more critical tasks, such as content refinement and student interaction. Additionally, offering comprehensive training on the new system will help ensure that staff feel confident in using AI-based tools.

An effective way to manage resistance is by involving employees in decision-making [8, p. 42]. By offering a gradual integration of the AI system, where manual processes are maintained alongside AI support, teachers and TAs may retain a sense of control over the workflow. This approach allows users to adapt to the new technology at their own pace, becoming more comfortable with AI over time. As a result, this strategy helps to reduce friction and increase the likelihood of successful adoption .

5.2.2 Security

The introduction of AI systems at NTNU raises security and data privacy concerns, particularly with regard to GDPR compliance. AI systems may rely on sensitive student data, such as academic performance and personal details, to function effectively. Mishandling this data can lead to break of privacy, putting both the institution and its students at risk.

Key GDPR principles must guide AI implementation at NTNU [8, p. 471]:

- **Purpose limitation**

AI systems should process student data solely for educational purposes, such as assignment creation and grading. Any further use requires explicit student consent.

- **Data minimization**

Only the necessary data should be collected for AI functions, minimizing the exposure of sensitive information.

- **Integrity and confidentiality**

Robust security measures, such as encryption and access controls, should be

in place to prevent unauthorized access and protect the integrity of student data.

- **Transparency**

Students must be fully informed about how their data is being used, with explicit consent obtained for processing.

By following these principles, NTNU can ensure that AI systems comply with GDPR while keeping the student data safe.

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